

Two-Branch Deepfake Detection Network Based on Improved Xception

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Abstract—With the continuous development of computer technology, the level of AI technology has also been greatly improved. But it comes with growing security and ethical challenges, including deepfake faces. While current detection methods work well in high definition video, they often do not perform as well when detecting relatively low definition video. For example, the clarity of the forged video is often too low, the quality compression of the video on the social platform and the level of the user's own equipment lead to the clarity is not too high, which makes the detection accuracy need to be improved. At the same time, the quality of detection across data sets also needs to be improved. To solve these problems, this paper proposes a two-branch detection network based on improved Xception. The network consists of a whole branch that detects the whole detected video and a local branch that detects each frame of the video. The whole branch uses the improved Xception network and Gated Recurrent Unit (GRU) to detect the whole video to be detected. The local branch uses the improved Xception network and a series of data enhancement measures, including Face-Cutout, to detect every frame of the detected video. To verify the effectiveness of the algorithm, tests were performed on the FF+ dataset and the Celeb-DF dataset. The experimental results show that the detection level of the proposed method is better than other detection networks, and it has better detection performance in cross-dataset and cross-definition scenarios, which proves the effectiveness of the proposed method.

Keywords—deepfake, deep learning, attention, Xception, forgery detection

I. INTRODUCTION

With the heat of deep learning in recent years, AI-based face-to-face technologies, such as Deep Fake, Face2Face[1], and Face Swap[2], are also rapidly evolving. These AI-based face-to-face technologies often reach the point where it is

difficult to distinguish the truth with the naked eye, poses many challenges to public safety and ethics. At present, AI face detection research has developed and has made great progress, of course there are some research difficulties. First of all, for low-definition forged face detection is not accurate enough, most face forged detection methods only in higher-defined images and videos have achieved better results, ignoring the study of low-discretion detection. In the actual situation, due to falsified material itself low clarity and the falsification faces producer's own lower level of equipment, the video that needs to be detected is often less clearness, which puts the cross-clarity capability of the detection method to the test. The second is the poor generalization capability between different databases. The detection methods for this category are not perfect, but AI face-to-face technologies, including DeepFake, Face2Face, and Face Swap, are evolving, and most of the methods tend to be trained for a particular data set of facial technology, which may not be accurately identified for the unknown AI face to-face technology, causing the model to be less generalized in practical applications.

In order to better address these difficulties, this paper proposes a two-branch depth-to-face detection network, which combines the features of video detection and image detection methods. In addition, improvements were made to Xception[3] through the introduction of the Convolutional Block Attention Module[4] to enhance its trait extraction capability, the extraction of time sequence information through the Gated Recurrent Unit[5] to improve detection capabilities, and a series of data enhancements to the dataset to improve the trait extract capability of the model. The experimental results showed that in the open datasets Celeb[6] and FaceForensics++[7], the method obtained better indicators than other methods, proving the method's effectiveness.

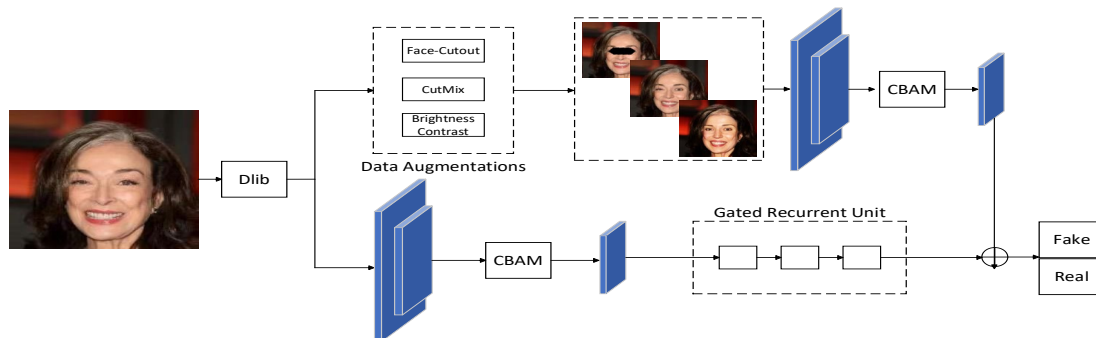


Fig.1. The structure of proposed work

II. PROPOSED WORK

Figure 1 depicts the two-branch network structure of this article. After receiving the material to be detected, the dlib face recognition module[8] extracts the facial features in the material and provides them to the video detection and frame detection branches, respectively. When the video branch receives the passed features, the Xception network enhanced by the attention mechanism will perform feature extraction and pass the features to the GRU network for timing information detection to realize video detection. To realize the detection of each frame of the video, when the frame detection branch gets the sent features, it performs a number of data processing procedures such as Face-Cutout[9] and other data enhancement strategies such as cut and splice before displaying the improved Xception. And then the whole system will hybrid the results of the two branches to output the detection results of the entire network.

A. Xception with Convolutional Block Attention Module

Xception is a neural network obtained by improving Inception v3. When traditional convolution is operated, it needs to be operated in the input channel dimension and spatial dimension at the same time, which leads to high coupling between cross-channel correlation and spatial correlation, whereas Xception adopts Depthwise separable convolution to replace the traditional convolution module, and improves the performance of the network by using the model parameters more effectively.

Depthwise Separable Convolution is a commonly used convolution operation in convolutional neural networks. It decomposes the standard convolution operation into two steps: Depthwise Convolution and Pointwise Convolution. In these two processes, it uses only one convolution kernel to convolve the input channels channel-by-channel, and uses a 1×1 convolution kernel for inter-channel blending, which reduces the computational complexity[10]. And although the amount of parameters is reduced, the depth separable convolution combines the information of different channels by point-by-point convolution, so that its own feature extraction ability is maintained. In order to improve the feature extraction ability, this paper uses CBAM to improve it, and the improved structure is shown in the Figure 2.

Attention mechanisms greatly enhance the performance of many computer visual tasks, while CBAM(Convolutional Block Attention Module) is a lightweight module that combines channel attention and spatial attention, and the CBAM module is able to sequentially generate attention feature information on the two dimensions of the channel and space, and then the two feature information is multiplied with the original input feature to make adaptation feature correction to produce the final feature picture. It has good applicability and can be embedded in almost any neural network to enhance model performance.

In order to improve the ability of Xception, CBAM module is used to the middle flow of the original Xception. It consists of eight separated convolutional blocks within a single branch, in order to improve the feature extraction capability of the network, the network is first divided into three parallel branches and an identical convolutional block is added, then each branch is equipped with two, three and four convolutional blocks, and a CBAM module is introduced after each branch in order to enhance the network's ability to detect features of different dimensions, and finally these

branches are re-combined using 1×1 convolutional blocks and reducing the dimensions of the delivered features. Finally these branches are re-merged using 1×1 convolutional blocks and the dimensionality of the delivered features is reduced.

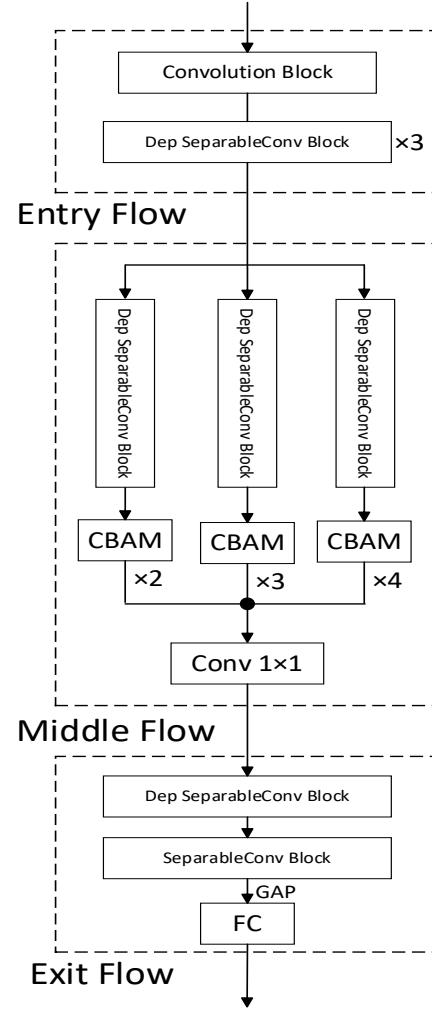


Fig.2. The structure of the improved Xception.

B. Gated Recurrent Unit

Vanishing Gradient is a very important problem in deep learning. That is, in a deeper network, the calculation of gradient will be close to 0 as the number of layers increases, resulting in the network parameters can not be updated. GRU neural network was initially designed to solve this problem, which is a variant of LSTM, based on LSTM neural network optimized structure reduces the parameters and speeds up the training speed. To solve the problem of Vanishing Gradient, GRU proposes to use two vectors, the update gate and the reset gate. The reset gate controls the mixing degree of the current input and the previous status information, so as to control the retention of historical information; The update gate controls the proportion of the current state to the previous state to control the integration of new information. It can save the information from long time ago and also removes the irrelevant information. In this method GRU network is used in video branching to receive spatial features passed from higher level network and retain them to extract inter-frame discontinuity information to improve detection.

C. Data Augmentations

1) Face-Cutout

The method is used to enhance data by erasing images, which is different from random erasing by using the random erasure method, by selecting rectangles of various sizes and replacing pixels with random values, although this enhancement is very effective in reducing over-adaptation. However, in experiments, it has been found that pure random wiping may have a negative effect on depth falsification detection, as it does not take into account the underlying pixel information of the deleted section, but only randomly deletes it. To improve the efficiency of the wipe, and to use it for depth forgery detections, the method provides selective wipe by masking areas of the face that do not contain falsified information.

2) CutMix

Regional dropout algorithms have been proposed to be effective in directing the model to attend to less discriminative aspects of objects. However, remove useful pixels from training images by superimposing a patch of either black pixels or random noise perhaps results in knowledge loss and inefficiency during training.

The main idea of CutMix[11] is to randomly crop a part of the region of the image, and then fill the cropped region

into the corresponding region of another image, which in turn achieves the effect of expanding the data samples, and then assign the classification labels proportionally to the mixing of fake and real images, so that the labels of the data are not only for the real and fake categories, which enhances the detection ability of the training samples.

3) Random Brightness Contrast

In computer vision, optical transformation-based data enhancement approaches are extensively utilized, and effective data improvements can help models to learn more robust features. According to , A single optical transformation method has limited performance improvements for depth forging detection models, and the impact on depth falsification detection model is not apparent, and some methods can even result in a decrease in performance. It is speculated that because of the small sample diversity generated by a single optic transformation data method, the network still focuses on a single forging feature and lacks data diversity. However, Random Brightness Contrast, an optical transformation approach used to modify the image's contrast and brightness, which blocks the underlying proofing characteristics in some false images, causing the model to learn to more robustly falsify hints.

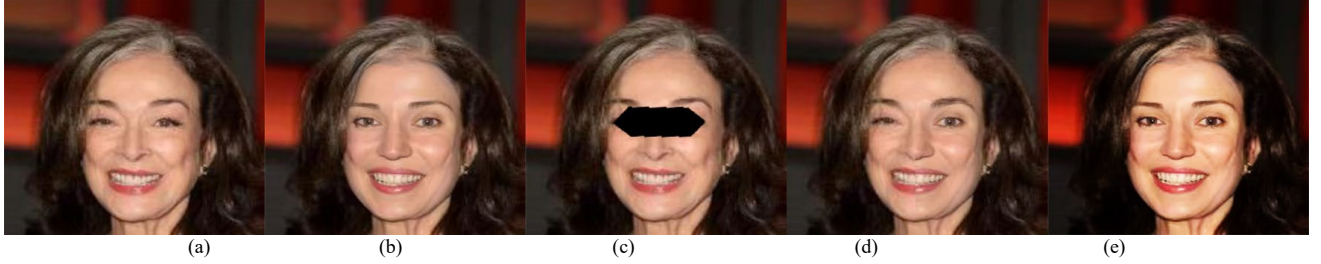


Fig.3. Data Augmentations used in proposed work

Figure.3 shows the image after the measures above.

III. EXPERIMENTAL SETUP

To improve the detection performance, the algorithms used the migration learning strategy, i.e. the ImageNet pre-training model. The learning rate adjustment strategy is set to Adam. The superparameter set to: learning rate to 0.0001, Weight Decay to 0.001, Batchsize to 40, epoch to 15. And to ensure the stability of the experimental results, the same random seeds are set for each experiment.

This method uses two branches to be tested separately, and the final result of the combination of the two Branches as the final score, the calculation formula is as follows.

$$P = KF_V + (1-K)F_F \quad (1)$$

Among them: P is the final prediction result; F_V is the prediction of the video branch stream; F_F is the forecast value for the brass branch; K is the balance factor, 0.5 is taken in experiments.

A. Experimental Environment Configuration

The experimental platform for this paper is a 64-bit Linux operating system, version number is Ubuntu 18.04 LTS, GPU is NVIDIA GeForce RTX 3090, with Intel Core i9-9900K, processor. Anaconda version 2020.11, code based on Python language, implemented under Pytorch framework for version 1.9.0.

B. Experimental Data

FF++ is one of the most popular video databases, which was launched in 2019 and is an extension of the original FaceForensics[12] database. The data sets consist of four manipulation methods. Each manipulation has 1,000 videos, plus the original video, and a total of 5,000 videos. To prove the validity of this algorithm, use the Deepfakes datasets in FF++ to use two compressed videos, c23 and c40. First, the data sets are divided into two groups based on official documents, Real and Fake, and each group is divided in corresponding training, testing and verification sets. After that, 10 frames are intercepted for each video, and the Dlib Face Recognition module is used to retrieve and save the face frames.

C. Evaluation Index

ACC and AUC were used as evaluation indicators in experiments. The calculation method for ACC is as follows:

$$Acc = \frac{TP+TN}{TP+TN+FP+FN} \quad (2)$$

Among them, True Positive (TP) is the algorithm for judging depth False images are depth false images false positive (FP) for judgment of real face images False face images, False Negative (FN) for Judgement of depth Fake images for real face pictures, True Negative for Judging Algorithm (TN) Real face images for true face images.

AUC is calculated as

$$AUC = \frac{\sum pred_{pos} > pred_{neg}}{posNum \times negNum} \quad (3)$$

In the formula: posNum is the number of positive samples, negNum the negative sample number, then the scale represents the combination of positive negative samples total; pred_{pos} represents a model's prediction of positive sample, pred_{neg} is the predictions of the model for negative samples, and molecules represent the combinations of the right sample greater than the negative, which means that each randomly extract a sample from the data set, the prediction value of the positive specimen is higher than the probability of the negative specimen.

IV. EXPERIMENTAL RESULTS

A. Detect Results and Comparison

In order to verify the effectiveness of this method, several common test methods are selected as comparison experiments. EfficientNet[13], Xception and ResNet, were chosen first for comparison. Furthermore, DSP-FWA[14], which is based on FWA, includes space pyramids to deal with varying input sizes of face-to-face images, and Mesonet[15], which captures the medial properties of face-to-face images to determine image authenticity, are both utilized to contrast.

Table I shows the result of every method.

TABLE I. TEST RESULTS ON FF++ DATASET

Methods	C23		C40	
	Acc(%)	AUC(%)	Acc(%)	AUC(%)
Xception	86.43	88.75	75.50	80.57
EfficientNet	88.32	89.28	76.16	82.17
ResNet	86.76	88.92	75.62	81.51
DSP-FWA	88.67	89.64	78.46	83.25
Mesonet	85.34	84.42	74.34	79.42
Proposed Method	89.64	90.12	80.64	86.53

From the above table, it can be seen that most character extraction networks can good detection results, in terms of performance, this method ACC is four to five percent higher than other methods, and the model also performs well in the AUC value.

The results of the cross-compression rate detection which is shown in Table II further illustrate the effectiveness of this method, in the case of different degrees of decrease in the accuracy of other models, the present model can still maintain precision, confirming the model's capacity to detect cross compression rate, while the other models' decreased accurateness also from the side proves that cross-compression is still a key problem in the detection of fake videos.

TABLE II. TEST RESULTS ON CELEB DATASET

Methods	AUC(%)
Xception	75.84
EfficientNet	77.78
ResNet	76.43
DSP-FWA	78.03
Mesonet	73.64
Proposed Method	82.37

B. Ablation Experimental Results

This part is ablation experiment on the FF++ data set to demonstrate the effectiveness of the modules in improving the detection effect. In order to conduct comparative experiments on different modules, this paper divides four

different experimental schemes as shown in the table below:
- Video correspondence of the detection branch of mercury, -
Frame corresponding to the video detection branches, -
CBAM representing removal of the CBAM attention module added to the Xception network, -
Augmentations representing the data enhancement process of removal in the frame detection branch, and comparing the above results with the base network Xception and the overall network structure proposed in this paper. And the results is in Table III.

TABLE III. ABLATION EXPERIMENTAL RESULTS

Methods	C23		C40	
	Acc(%)	AUC(%)	Acc(%)	AUC(%)
-Video	87.59	89.32	78.46	84.64
-Frame	87.61	89.46	79.34	83.76
-CBAM	88.49	89.84	79.76	85.46
-Augmentations	88.61	89.73	79.84	85.71
Xception	86.43	88.75	75.50	80.57
Proposed Method	89.64	90.12	80.64	86.53

This paper presents a network structure that is superior in performance to each part of the network, and analyzes the reasons for this as follows: the false face images obtained by the forgery technology are compressed, the tampering traces in the image will be contaminated by the compression errors, the characteristic information captured only by the image or video channel will be significantly reduced, and the two-branch network structure of this article is better resulted by extracting space and timeline information simultaneously through the video branch and the forging branch, thereby enhancing each other. And the data enhancement method used in this paper allows the neural network to avoid learning about the local low-level characteristics of the forged image, instead of learning more advanced characteristics, improved the detection capacity of the model. Because falsified facial images often contain both real and manipulated areas, the introduction of CBAM attention mechanisms to the detection network allows the network model to allocate more attention to the manipulation area, which significantly improves the performance of detection.

V. CONCLUSION

In response to the problem of low accuracy detection in low-quality video sets and cross-archive scenarios that exists in the existing deep false detection methods, this paper proposes a two-branch depth false Detection method, constructing video and single-branch double-branches false face detection network structure to detect spatial information and timescale information, while using the CBAM module to improve the Xception network, and use different data enhancement methods to process the data. Comparison experiments in different datasets show that the network model presented in this paper has good detection performance and generalization capabilities, and is superior to existing detection methods, while melting experiments conducted for each module prove the effectiveness of the improvement measures in this article.

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